

FPGA-Based Multi-Sensor Monitoring System with MicroBlaze Processor

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Abstract— This project implements a comprehensive environmental monitoring system utilizing FPGA technology. Developed on the Digilent Nexys A7-100T platform with a Xilinx Artix-7 FPGA, the system integrates multiple sensor modules including ambient light, temperature, and distance detection. A MicroBlaze soft processor orchestrates data acquisition from Pmod peripheral modules while driving an RGB OLED display for real-time visualization. The design exemplifies embedded systems engineering principles through its modular architecture, hardware-software co-design methodology, and systematic validation approach.

Keywords—

Group member Contributions	
Jad Al Jourdi	Created block designs for all pmods individually and combined on vivado. Made vitis workspaces for all pmods individually and combined (OLEDrgb + ALS + MAXSONAR)
Raihan Uddin	Implementation of TMP3 and it's pin mapping
Raphael Catchpole	Documentation

I. PROJECT OVERVIEW

A. Key Features

The system integrates multiple environmental sensors measuring and monitoring ambient light, temperate and distance allowing for comprehensive collection of data. The real-time visualization is achieved through an OLED. A USB-UART interface is used for Communication and debugging with a PC.

Hardware architecture is modular and supports incremental sensor integration and verification.

Pre-configured FPGA bitstreams are used for testing and deployment across different Hardware configurations.

B. Development Platforms

The Target board is Digilent Nexys A7 using an Artix-7 FPGA. The hardware was designed in Xilinx Vivado 2021.1 and Vitis 2021.1 was used for writing the embedded software.

An Arduino-compatible setup was used to test inter-sensory communication before moving on tot he full FPGA system.

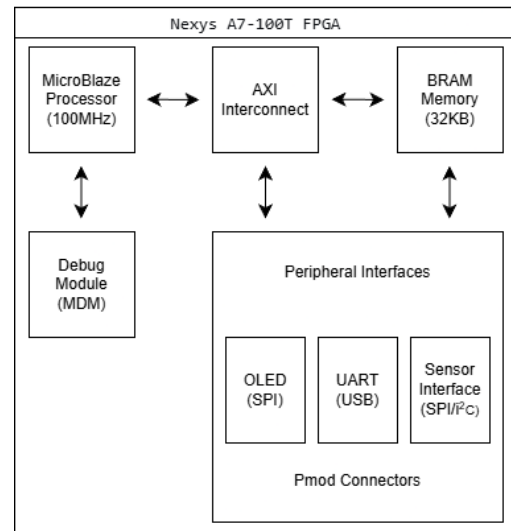


Fig. 1. Block Diagram Overview

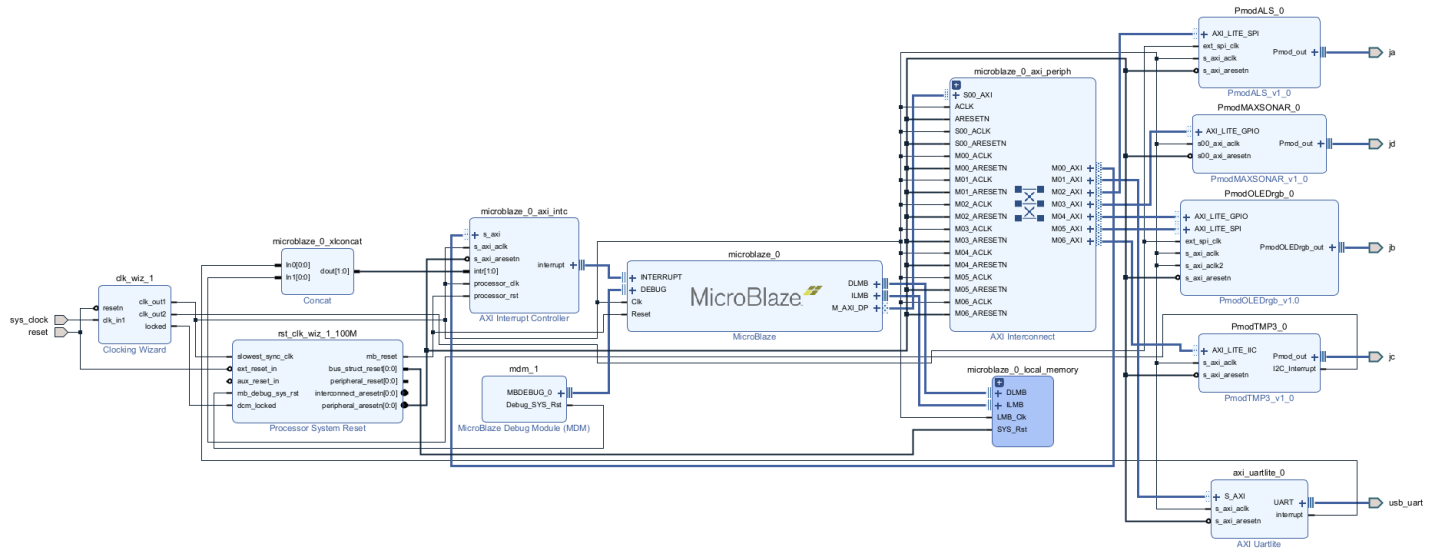


Fig. 2. MicroBlaze-based embedded system showing processor, AXI interconnect, peripheral modules, and control components.

